

ELEFSIS, Greece

WMO: 167180

Lat: 38.064N

Lon: 23.556E

Elev: 143

StdP: 14.62

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.5	36.0	19.6	14.8	39.0	24.4	18.6	41.2	28.3	44.9	23.2	45.8	7.6	0	0.438

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	98.6	69.2	95.6	68.9	93.4	68.3	72.8	88.8	71.8	88.3	70.7	87.6	11.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.1	103.8	79.7	66.3	97.5	79.0	64.6	91.8	78.7	36.6	88.4	35.6	87.9	34.6	87.5	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.7	19.0	17.4	29.1	105.0	3.3	4.0	26.8	107.8	24.9	110.2	23.0	112.4	20.7	115.3
			25.9	75.3	3.4	1.3	23.5	76.2	21.6	77.0	19.7	77.8	17.2	78.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.2	48.9	49.8	53.8	60.2	69.7	78.6	83.8	83.8	75.8	67.0	58.7	51.4
	DBStd	13.60	5.65	5.69	5.42	5.03	5.13	5.24	3.78	3.91	4.94	5.49	5.45	5.82
	HDD50	240	86	66	26	2	0	0	0	0	0	0	6	54
	HDD65	2108	499	426	349	155	15	0	0	0	1	43	198	422
	CDD50	5790	52	60	143	309	610	858	1046	1048	774	527	266	97
	CDD65	2183	0	0	1	12	160	408	581	583	325	104	9	0
	CDH74	23515	0	0	7	110	1240	4384	7359	7216	2694	482	23	0
	CDH80	10891	0	0	0	16	336	1996	3817	3669	983	73	1	0
Wind	WSAvg	6.9	6.7	6.7	6.6	6.3	6.7	7.4	8.2	8.6	7.0	6.7	5.7	6.4
Precipitation	PrecAvg	16.7	2.3	1.9	1.8	1.0	0.7	0.4	0.3	0.3	0.9	1.7	2.6	3.0
	PrecMax	25.9	5.0	5.5	5.8	2.4	2.2	1.4	1.3	1.1	4.7	5.3	6.0	7.2
	PrecMin	8.6	0.2	0.4	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.4
	PrecStd	4.3	1.4	1.3	1.4	0.7	0.6	0.4	0.4	0.3	1.1	1.3	1.6	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.1	68.0	73.7	82.4	91.5	100.3	103.6	102.0	95.1	86.0	77.1	69.7
		MCWB	55.7	55.7	56.6	59.7	65.4	67.7	69.7	69.3	67.5	65.5	62.9	60.1
	2%	DB	62.8	64.5	69.8	76.9	86.4	95.4	98.6	98.5	91.5	82.0	73.3	66.2
		MCWB	54.7	54.2	56.2	59.3	63.7	68.4	69.3	69.5	67.7	64.9	62.6	58.1
	5%	DB	60.8	62.3	66.4	73.6	83.9	93.0	96.0	95.3	88.2	78.9	70.2	64.0
		MCWB	54.0	53.9	54.6	58.0	63.0	67.7	69.1	69.2	67.0	64.5	61.2	56.6
	10%	DB	58.8	59.4	64.1	71.2	80.7	89.7	93.4	93.2	85.9	76.7	68.0	60.9
		MCWB	52.0	52.2	53.9	57.1	62.1	67.1	68.4	68.6	66.4	63.9	59.9	54.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.3	58.1	59.4	63.3	67.7	72.0	74.2	75.0	72.9	70.5	66.9	62.2
		MCDB	63.8	65.0	69.1	75.7	83.6	89.9	88.7	88.8	84.5	78.0	73.3	67.3
	2%	WB	56.1	56.1	57.6	60.9	66.3	70.7	72.7	73.1	71.0	68.8	64.2	59.6
		MCDB	61.1	62.0	66.0	73.0	82.1	88.9	89.3	89.6	82.5	76.9	70.3	64.7
	5%	WB	54.4	54.6	56.3	59.4	64.8	69.6	71.7	71.9	69.7	67.0	62.5	57.4
		MCDB	59.6	60.6	64.0	70.8	79.5	87.9	89.4	88.9	82.0	75.5	68.7	62.0
	10%	WB	52.5	52.9	54.8	57.9	63.5	68.5	70.6	70.6	68.4	65.2	60.5	55.2
		MCDB	57.7	58.9	62.4	68.8	77.4	86.3	89.0	88.3	81.7	73.7	66.8	60.2
Mean Daily Temperature Range	5% DB	MDBR	13.5	14.4	16.0	17.8	18.9	19.1	18.3	17.5	16.8	14.8	13.9	13.1
		MCDBR	17.1	18.4	20.6	22.6	22.2	22.3	21.7	20.3	19.9	18.1	17.1	16.8
		MCWBR	11.0	11.3	11.0	10.9	9.4	8.5	8.7	8.5	8.8	8.8	9.9	11.0
	5% WB	MCDBR	15.1	16.6	18.2	20.5	20.4	20.0	19.7	18.7	18.5	16.4	15.1	14.5
		MCWBR	10.6	11.2	11.0	10.9	9.7	8.9	8.8	8.9	9.1	9.2	9.6	10.4
Clear-Sky Solar Irradiance	taub	0.355	0.380	0.418	0.463	0.445	0.425	0.429	0.436	0.432	0.404	0.382	0.365	
	taud	2.408	2.317	2.207	2.092	2.207	2.303	2.301	2.272	2.278	2.332	2.357	2.378	
	Ebn at Noon	258	266	266	260	267	272	270	265	259	254	245	244	
	Edh at Noon	28	34	42	50	45	41	41	41	39	33	29	27	
All-Sky Solar Radiation	RadAvg	623	868	1298	1718	2059	2358	2402	2138	1615	1092	730	540	
	RadStd	74	106	108	153	105	122	56	50	92	95	76	75	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.90	N/A	N/A	+0.99	N/A	N/A	N/A	-153	N/A	N/A
(47)	Regional (12 neighbors)	+0.54	N/A	N/A	N/A	N/A	N/A	N/A	-130	+153	N/A

Nomenclature: See separate page